OPTIONAL CHALLENGE – EXTRA PIPELINE STAGE

1. Design Approach:

Designed a six-stage pipelined processing unit using Verilog HDL to demonstrate data processing across multiple pipeline stages. Implemented addition, subtraction, bitwise AND and bitwise OR operations. Used valid signals to indicate data movement between pipeline stages and to identify when the output result is available.

2. Verification Steps:

- Created a separate testbench.
- Applied reset to initialize the pipeline.
- Provided sample input operands and operation codes.
- Tested addition, subtraction, AND and OR operations.
- Checked valid signal propagation through pipeline stages.
- Verified output results against expected values.
- Simulated using ModelSim.

3. Results:

The testbench checks the execution of four ALU operations through the pipeline. Expected outputs are 15, 20, 8 and 15 respectively. PASS/FAIL messages are displayed for output comparison.

4. Issues Faced:

Maintaining correct data transfer between pipeline stages was challenging. Valid signal synchronization is important to identify the correct output cycle. Proper handling of arithmetic operations is also required.

5. Conclusion:

Successfully designed a six-stage pipeline demonstration module using Verilog HDL to perform and monitor different ALU operations.